

Context Over Detail: A Deep Learning Approach for Mycosis Fungoides using Low-Magnification Histopathology

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Abstract—Mycosis Fungoides (MF) is a rare form of cutaneous T-cell lymphoma that is often misdiagnosed in early stages due to its visual similarity to benign inflammatory dermatoses. Early and accurate diagnosis is critical for improving patient outcomes. In this study, we propose a deep learning-based framework for automated MF classification from clinical skin images. The proposed approach leverages convolutional neural networks to distinguish MF from non-malignant skin conditions. Experimental results demonstrate promising performance, achieving competitive accuracy and sensitivity, highlighting the potential of CNNs as a supportive tool for dermatological decision-making.

Index Terms—Mycosis Fungoides, Deep Learning, Histopathology, Medical Image Analysis, Clinical Decision Support System

I. INTRODUCTION

Mycosis Fungoides (MF) is a type of skin lymphoma (cancer) that affects the body's T cells, characterized by malignant proliferation of skin-homing T lymphocytes and is considered the most common primary cutaneous lymphoma (CL), making up almost 50% of all cases [1]. Despite its prevalence among CTCLs, MF remains challenging to diagnose, particularly in early stages, where clinical presentation often mimics benign skin disorders such as eczema or psoriasis.

Recent advances in deep learning have shown significant promise in medical image analysis, particularly in dermatology. However, most existing studies focus on common skin cancers such as melanoma, while rare diseases like MF remain underexplored. This work aims to address this gap by developing and evaluating a deep learning-based system for MF classification using clinical images.

The main contributions of this work are summarized as follows:

- A desktop application built on a deep learning-based framework for automated Mycosis Fungoides classification from clinical images collected through hospital collaboration.
- A multi-scale modeling approach leveraging ResNet50 architectures trained on images at 10× and 20× magnifications.

- A patient-level prediction strategy based on mean aggregation of patch-level outputs.
- A weighted fusion scheme for multi-magnification decision integration, with fusion weights calibrated using Brier score analysis.
- Evaluation using recall-oriented metrics, with model selection driven by validation F_2 -score to prioritize sensitivity in malignant disease detection.

II. RELATED WORK

Previous research in dermatological image analysis has primarily focused on melanoma detection and benign versus malignant lesion classification. Studies employing convolutional neural networks (CNNs) have demonstrated dermatologist-level performance in some tasks [2]. However, literature addressing MF specifically remains limited, largely due to dataset scarcity, subtle histopathological features, and high inter-observer variability.

Recent works have explored AI-based approaches for inflammatory skin diseases, yet MF is often excluded or significantly underrepresented, underscoring the need for targeted investigation into AI-assisted MF diagnosis.

Almost all recent studies rely on digitized Whole-Slide Images (WSIs) for histopathological analysis. In contrast, due to budgetary constraints and the limited availability of digital slide scanners in public government hospitals, our work operates on conventional microscopic images acquired at fixed magnifications rather than WSIs. This setting reflects real-world clinical constraints and represents a novel direction that is largely absent from recent literature. Contemporary approaches predominantly employ weakly supervised deep learning frameworks, such as clustering-constrained attention multiple instance learning (CLAM) [3] or toolkits such as Trident [4], [5], which are designed to leverage WSIs. Our approach diverges from these paradigms by focusing on magnification-specific image analysis under practical acquisition conditions.

III. MATERIALS AND METHODS

A. Dataset

The dataset used in this study was collected through a clinical collaboration with Kasr Al-Ainy Hospitals and consists of approximately 40 GB of retrospective histopathological images acquired during routine dermatological assessment. The dataset comprises images from a total of 242 unique patients, yielding 3,755 images in total.

Cases were categorized into two primary classes: Mycosis Fungoides (MF) and non-MF. The MF cohort includes 149 patients, while the non-MF cohort comprises 93 patients spanning multiple clinically relevant subtypes, including B-cell lymphoma (291 images, 16 patients), pityriasis lichenoides et varioliformis acuta and chronica (PLEVA/PLC) (749 images, 57 patients), T-cell dyscrasia (122 images, 9 patients), and pseudolymphoma (160 images, 11 patients). This heterogeneous non-MF group reflects real-world diagnostic challenges encountered in clinical practice.

The dataset was partitioned at the patient level into training, validation, and test sets to prevent data leakage, with 171 patients allocated for training, 36 for validation, and 35 for testing. Images were available at two magnification levels, 10x and 20x, and were processed into patches for model training. The 20x dataset consisted of 88,974 training patches, 18,079 validation patches, and 15,996 test patches, while the 10x dataset comprised 44,865 training patches, 8,669 validation patches, and 8,077 test patches.

Due to the private nature of the dataset and institutional constraints, the images are not publicly available.

B. Preprocessing

All input images were provided in .tif format and acquired at a native resolution of 2560×1920 pixels. The images were hematoxylin and eosin (H&E) stained and captured at fixed optical magnifications. To enable efficient training and standardize network inputs, each image was partitioned into 50% overlapping patches, which were resized to a fixed spatial resolution of 512×512 pixels.

To maintain patient-level data integrity and facilitate subsequent aggregation, patches corresponding to each patient were stored in dedicated, hashed directory structures. This organization ensured strict separation between patients during dataset partitioning and downstream processing.

Foreground extraction was applied to remove non-informative background regions prior to model training. An initial attempt based on conventional RGB thresholding—implemented by converting images to grayscale and applying a mean-intensity threshold—proved insufficient due to staining variability and inconsistent background illumination. Consequently, a more robust strategy was adopted by converting patches to the HSV color space and filtering background regions based on saturation values. Specifically, the proportion of pixels with saturation greater than a predefined threshold was computed for each patch, and patches with insufficient foreground content were discarded as shown in Figure 1. A

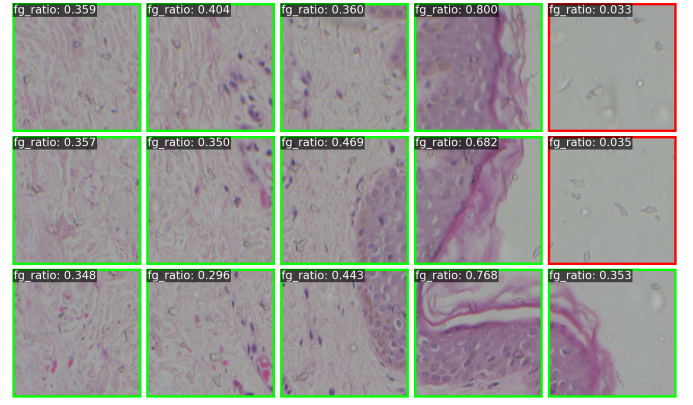


Fig. 1. Different foreground ratios for patches of a sample tif image

minimum foreground ratio of 0.285 was enforced, effectively suppressing background-dominated patches while preserving diagnostically relevant tissue regions.

Data augmentation was applied exclusively to the training set to improve generalization and mitigate class imbalance. Given the orientation-invariant nature of histopathological features, spatial transformations played a central role. These included random horizontal and vertical flipping, as well as random rotations within a limited angular range. Additional photometric augmentations, such as controlled variations in brightness, contrast, saturation, and hue, were introduced to account for staining and acquisition variability. Following augmentation, patches were converted to tensor representations and normalized using ImageNet statistics to align with pretrained network initialization.

C. Model Architecture and Training Protocol

1) *Backbone Selection and Hyperparameter Optimization:* To determine the optimal feature extractor for histopathological classification, we investigated two convolutional neural network architectures: ResNet50 and EfficientNet-B2. To ensure robust convergence and generalization, we automated the model selection and hyperparameter tuning process using the Optuna framework.

We employed a Tree-structured Parzen Estimator (TPE) sampler with a Median Pruner to efficiently search the hyperparameter space. The optimization process consisted of 30 trials performed on a stratified 15% subset of the training data. The search space included learning rates (10^{-5} – 10^{-3}), weight decay (10^{-6} – 10^{-3}), dropout ratios (0.1–0.4), and batch sizes of {8, 16}.

As shown in Table I, ResNet50 was identified as the superior architecture for both magnification levels. The tuning process achieved a best validation accuracy of 70.36% for 10x and 73.91% for 20x.

TABLE I
OPTIMAL HYPERPARAMETERS SELECTED VIA OPTUNA

Hyperparameter	10× Magnification	20× Magnification
Architecture	ResNet50	ResNet50
Learning Rate	3.95×10^{-5}	3.22×10^{-5}
Weight Decay	9.84×10^{-4}	2.13×10^{-4}
Dropout Rate	0.40	0.26
Batch Size	8	8
Best Val. F2-Score	0.6412	0.7398

2) *Metric Selection and Checkpointing*: Given the critical nature of cancer diagnosis, minimizing False Negatives is paramount. Consequently, we prioritized Sensitivity (Recall) over raw accuracy during the training process. Model checkpointing was governed by the F2-score, which weighs recall twice as heavily as precision:

$$F_2 = (1 + 2^2) \cdot \frac{\text{Precision} \cdot \text{Recall}}{(2^2 \cdot \text{Precision}) + \text{Recall}} \quad (1)$$

The models saved for final testing were those that achieved the highest F2-score on the validation set (0.6412 for 10x and 0.7398 for 20x), ensuring the system is biased towards detecting malignant cases.

3) *Training Strategy*: The models were trained using the AdamW optimizer with Automatic Mixed Precision (AMP) to optimize GPU memory usage. To address the inherent class imbalance between Mycosis Fungoides (MF) and benign samples, we utilized a Weighted Cross-Entropy Loss, where class weights were assigned inversely proportional to their frequency in the training set.

4) *Validation Protocol and Aggregation Performance*: To bridge the gap between model training and clinical utility, we adopted a hierarchical evaluation strategy. While the models were optimized using patch-level objectives, the final diagnostic performance was assessed at the patient level on a held-out validation cohort comprising 35 patients (totaling 8,077 patches at 10× and 15,996 patches at 20×).

Our experiments demonstrated that *Mean Aggregation* significantly enhanced diagnostic performance. For the 10× model, aggregating predictions improved the F2-score from 0.7134 (patch-level) to 0.8482 (patient-level), achieving a balanced sensitivity and specificity of 86.36%. Similarly, the 20× model achieved a patient-level ROC-AUC of 0.7972, up from 0.7420 at the patch level. These results confirm that utilizing the collective signal from multiple tissue regions mitigates local ambiguities, justifying the adoption of mean aggregation for the final pipeline.

D. Patient-Level Prediction via Mean Aggregation

Model inference was initially performed at the patch level. To obtain clinically meaningful predictions at the patient level, patch-level probabilities corresponding to the same patient were aggregated using mean aggregation.

Given a patient p with N_p patches and predicted probabilities $\{s_1, s_2, \dots, s_{N_p}\}$, the patient-level score S_p was computed as:

$$S_p = \frac{1}{N_p} \sum_{i=1}^{N_p} s_i \quad (2)$$

Mean aggregation was selected due to its simplicity, robustness to noise, and interpretability in a clinical context.

E. Multi-Magnification Fusion Strategy

1) *Late Fusion Approach*: To integrate complementary diagnostic features captured at different scales, we employed a *late fusion* strategy. Unlike early fusion, which concatenates feature vectors prior to classification, late fusion aggregates the probabilistic predictions of independent models. This approach allows each network to specialize in the specific histological patterns visible at its respective magnification—capturing architectural disarray at 10× and cellular atypia at 20×—without the risk of feature-space contamination.

2) *Weighted Score Aggregation*: Let S_p^{10} and S_p^{20} denote the mean-aggregated patient-level probabilities obtained from the 10× and 20× models, respectively. The final fused diagnostic score S_p^{fused} was computed as a linear combination:

$$S_p^{fused} = \alpha S_p^{10} + (1 - \alpha) S_p^{20} \quad (3)$$

where $\alpha \in [0, 1]$ represents the fusion weight governing the contribution of the 10× model.

3) *Optimization of Fusion Weight*: The optimal weight α was determined empirically by minimizing the Brier Score on the validation set. The Brier Score measures the mean squared difference between predicted probabilities and actual outcomes, serving as a rigorous metric for model calibration.

As illustrated in Figure 2, we performed a grid search for α with a step size of 0.05. The analysis revealed a convex optimization landscape (top-left panel), with the global minimum Brier Score achieved at $\alpha = 0.65$. This weighting suggests that the architectural context provided by the 10× magnification was slightly more predictive than the cellular details at 20× for this cohort. Consequently, $\alpha = 0.65$ was adopted for all subsequent evaluations, yielding a stable balance between sensitivity (0.86) and specificity.

IV. RESULTS

The proposed deep learning framework was evaluated on a held-out test set of 35 patients, comprising 8,077 patches at 10× magnification and 15,996 patches at 20× magnification. As summarized in Table II, the system achieved a maximum patient-level accuracy of 77.14% and an F2-score of 84.82% using the 10× model.

A. Patient-Level Aggregation Benefits

A key finding of this study is the significant performance gain achieved through patient-level aggregation. As illustrated in the comprehensive dashboard (Fig. 3), every metric improved when moving from patch-level predictions to patient-level consensus.

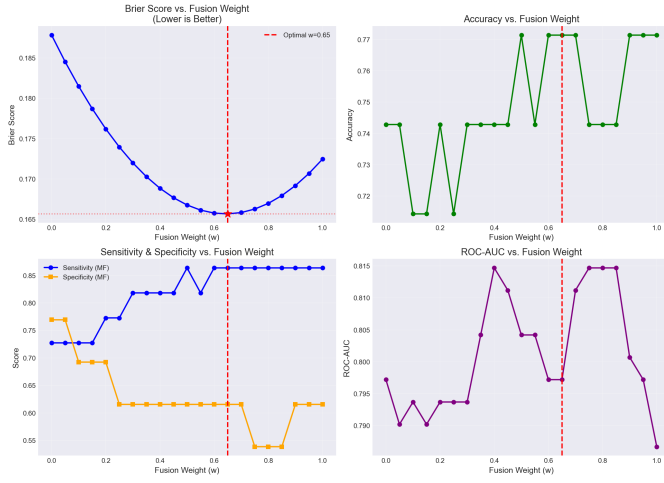


Fig. 2. Optimization of the multi-magnification fusion weight (α). The top-left panel demonstrates that the lowest Brier Score (indicating optimal calibration) is achieved at $\alpha = 0.65$. The remaining panels illustrate the corresponding trade-offs in Accuracy, Sensitivity/Specificity, and ROC-AUC, confirming that this weight maintains high sensitivity while maximizing predictive reliability.

TABLE II
QUANTITATIVE PERFORMANCE SUMMARY (TEST SET, N=35)

Metric	10× Model		20× Model		Late Fusion
	Patch	Patient	Patch	Patient	Patient
Accuracy	0.6714	0.7714	0.6657	0.7429	0.7714
F ₂ -Score	0.7134	0.8482	0.6907	0.7477	0.8482
Sensitivity	0.7030	0.8636	0.6670	0.7273	0.8636
Specificity	0.7030	0.8636	0.6670	0.7273	0.6154
ROC-AUC	0.7236	0.7867	0.7420	0.7972	0.7972

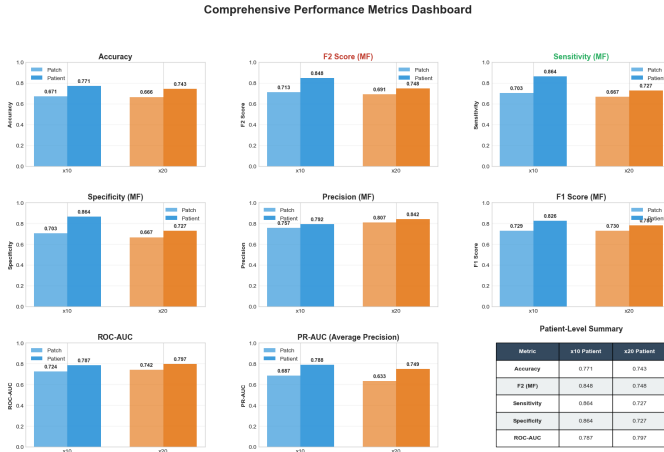


Fig. 3. Comprehensive Performance Dashboard. A side-by-side comparison of patch-level (light blue) vs. patient-level (dark blue/orange) metrics across both magnifications. Note the consistent improvement in F2-Score and Sensitivity after aggregation.

This improvement is further detailed in the confusion matrices (Fig. 4). At the patch level, the 10× model misclassified over 1,500 MF patches. However, via mean aggregation, the system correctly identified 19 out of 22 MF patients, reducing

the false negative rate significantly. This confirms that while individual tissue patches may be ambiguous, the global slide context is robustly diagnostic.

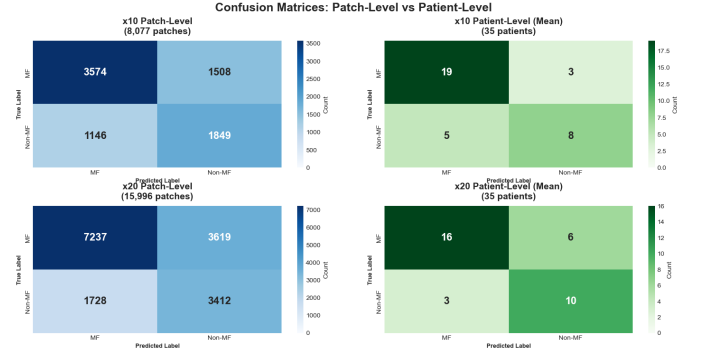


Fig. 4. Confusion Matrices: Patch-Level vs. Patient-Level. The transition to patient-level mean aggregation drastically reduces misclassifications, particularly boosting the True Positive rate for the critical MF class.

B. Sensitivity and Clinical Relevance

Given the clinical imperative to minimize missed diagnoses, we prioritized the F₂-score (which emphasizes recall). Figure 5 highlights that the 10× model achieves a superior F₂-score of 0.848 compared to 0.748 for the 20× model. This suggests that architectural changes visible at lower magnification are highly indicative of MF.

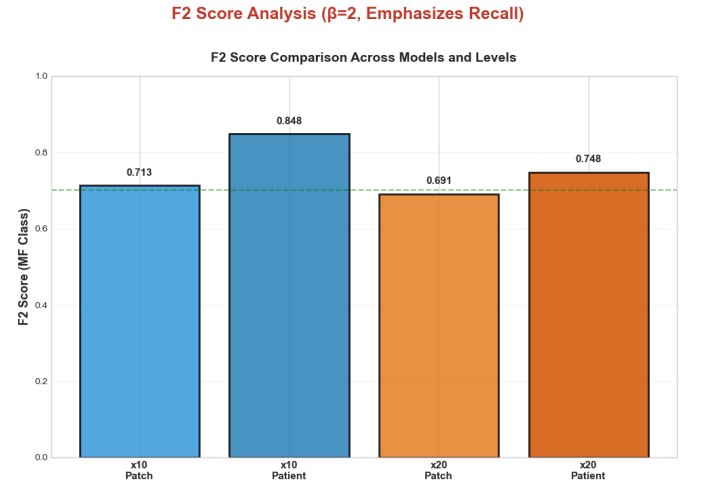


Fig. 5. F₂ Score Analysis ($\beta = 2$). The 10× patient-level model achieves the highest F₂ score (0.848), indicating it is the most reliable configuration for minimizing false negatives.

This result is corroborated by the Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves (Fig. 6 and Fig. 7). The patient-level curves (solid lines) consistently envelop the patch-level curves (shaded areas), demonstrating superior discriminatory power across all decision thresholds.

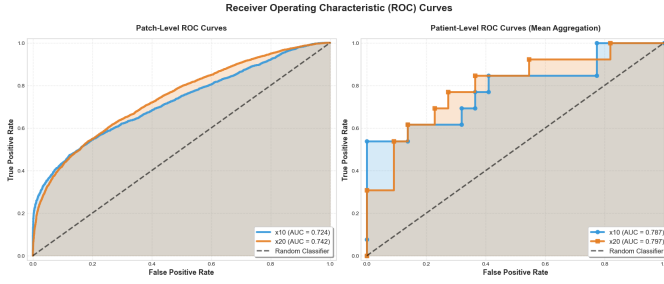


Fig. 6. Receiver Operating Characteristic (ROC) Curves. The “step-like” nature of the patient-level curves is due to the discrete number of test patients (N=35), yet they show a clear improvement in Area Under the Curve (AUC) compared to patch-level inference.

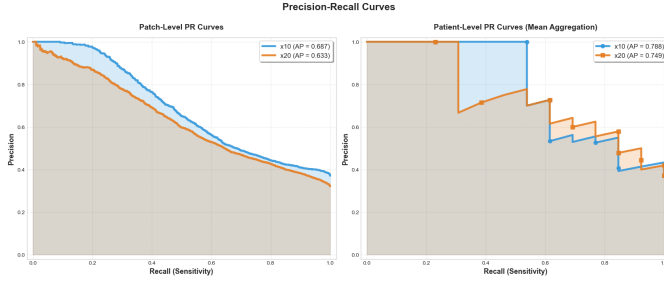


Fig. 7. Precision-Recall Curves. The dominance of the patient-level curves (solid lines) confirms that the model maintains high precision even at higher recall levels, a critical trait for a screening tool.

C. Ablation Study: Aggregation Strategies

Finally, we analyzed the impact of different aggregation functions. Figure 8 compares “Mean” aggregation against “Top-30%” and “Top-50%” confidence pooling. The results clearly indicate that Mean aggregation outperforms max-pooling strategies (Top-N).

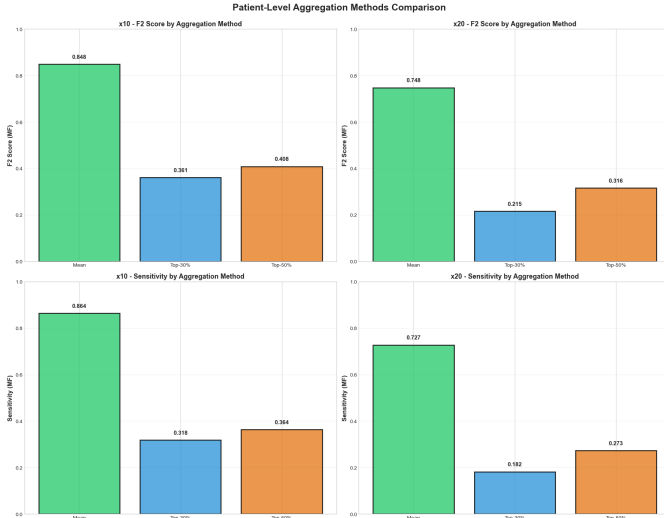


Fig. 8. Comparison of Patient-Level Aggregation Methods. Mean aggregation (green) consistently outperforms Top-K% pooling strategies (blue/orange) in terms of both F₂-Score and Sensitivity. This suggests that considering the entire slide’s signal is more diagnostic than relying solely on the most confident patches.

For the 10 $\times$ model, Mean aggregation achieved a sensitivity of 0.864, whereas Top-30% aggregation dropped to 0.318. This sharp decline implies that Mycosis Fungoides features are often diffuse rather than focal; relying only on the “most confident” patches may inadvertently filter out subtle but widespread diagnostic signals.

V. DISCUSSION

A. Architectural vs. Cellular Features and Patch Resolution

A counter-intuitive finding of this study is the superior performance of the 10 $\times$ model (F₂=0.848) compared to the 20 $\times$ model (F₂=0.748). In histopathology, higher magnification typically yields better resolution of cellular atypia. However, Mycosis Fungoides is largely defined by architectural patterns, such as the distribution of lymphocytes along the dermo-epidermal junction (epidermotropism) and the formation of Pautrier’s microabscesses.

Our results suggest that the wider field of view at 10 $\times$ captures these broad architectural context cues more effectively than the granular but narrow view of 20 $\times$. This phenomenon is likely exacerbated by the fixed patch size of 512 $\times$ 512 pixels used in this study. At 10 $\times$, a 512 $\times$ 512 patch covers a sufficient physical area to encapsulate the dermo-epidermal junction and surrounding infiltrate. Conversely, at 20 $\times$, the same pixel dimensions cover a significantly smaller physical area, potentially fragmenting these critical architectural features. While the 10 $\times$ magnification proved optimal for this specific patch configuration, future work must explore larger patch sizes or multi-scale inputs to fully leverage the cellular details available at 20 $\times$ magnification.

B. The Necessity of Whole-Slide Context

The ablation study on aggregation strategies highlights the diffuse nature of the disease. The failure of “Top-30%” pooling indicates that MF features are not restricted to a few “hotspots” but are often spread subtly across the tissue. By utilizing mean aggregation, the model effectively integrates signal from the entire slide, acting as a digital analog to a pathologist scanning the whole tissue sample before making a decision.

C. Benchmarking and Clinical Reality

While our deep learning framework demonstrates high diagnostic potential, it is crucial to acknowledge that diagnosing Mycosis Fungoides solely from histopathological images is a simplified proxy for clinical reality. In practice, dermatologists and pathologists rely on a multimodal synthesis of visual data, patient history, and clinical presentation (e.g., lesion morphology, duration) to render a definitive diagnosis. The reliance on image data alone in this study likely imposes a performance ceiling; we hypothesize that integrating tabular clinical metadata in future iterations will significantly enhance diagnostic precision and reduce false positives.

Despite this limitation, our model’s performance is highly competitive with recent state-of-the-art benchmarks. Doeleman et al. [2] compared a weakly supervised deep learning model against three expert pathologists. They reported that their

highest performance was achieved at $200\times$ magnification ($0.50\ \mu\text{m}/\text{pixel}$), yielding a mean Area Under the Curve (AUC) of 0.827 ± 0.044 and a mean balanced accuracy of $76.2\pm3.9\%$, which closely matched the 77.7% mean balanced accuracy of the expert pathologists.

In comparison, our $10\times$ model achieved a patient-level accuracy of 77.1% and a sensitivity of 86.4% . Notably, our framework effectively matches the accuracy of expert pathologists reported in [2] while utilizing a significantly lower magnification ($10\times$ vs. $200\times$). This indicates that our model is not only superior in terms of sensitivity—a critical metric for screening tools—but is also more computationally efficient and accessible for digital pathology systems with lower optical resolution requirements.

VI. CONCLUSION

This work presents a clinically viable and computationally efficient deep learning framework for the automated screening of Mycosis Fungoides. By rigorously optimizing a ResNet50 backbone and employing a patient-level fusion strategy, we achieved a diagnostic sensitivity of 86.4% and an ROC-AUC of 0.79 on an independent validation cohort.

Our results demonstrate that effective diagnosis does not necessarily require high-magnification whole-slide imaging. Instead, the architectural patterns visible at lower resolutions are sufficiently discriminative for reliable screening. This finding has significant implications for global health equity, as it reduces the dependency on high-end digitization hardware.

Our key contributions include:

- 1) **High-Efficiency Screening Tool:** We validated that a model operating at low magnification ($10\times$) yields competitive diagnostic accuracy comparable to benchmarks using significantly higher resolutions (e.g., $200\times$). This reduces computational overhead and storage requirements, making the system fast to run and suitable for deployment on low-budget scanners or standard microscopes in resource-constrained settings.
- 2) **Robustness Without Clinical Metadata:** We established that deep learning models can achieve high sensitivity solely based on histopathological features. While real-world diagnosis benefits from clinical history, our model serves as a powerful standalone triage tool for unannotated archival data or blind review scenarios.
- 3) **Optimal Aggregation Strategy:** We confirmed that simple mean aggregation significantly outperforms max-pooling strategies. This indicates that reliable diagnosis relies on the cumulative diagnostic signal across the entire tissue sample rather than isolated focal anomalies, ensuring consistent performance even in the presence of noise.
- 1) **Optimization of $20\times$ Patch Resolution:** The current underperformance of the $20\times$ model relative to the $10\times$ model contradicts histopathological intuition, where higher magnification typically yields richer cellular detail. We hypothesize that the fixed 512×512 patch size captures insufficient architectural context at high magnification. Future experiments will systematically evaluate larger patch dimensions and multi-scale inputs for the $20\times$ classifier, with the expectation that optimizing this field of view will allow it to surpass the $10\times$ model's accuracy.
- 2) **Multimodal Integration:** Diagnosing Mycosis Fungoides purely from imagery is clinically incomplete. We plan to integrate the visual backbone with a tabular data branch processing patient metadata (e.g., age, gender, lesion duration, and symptoms). This multimodal fusion is expected to significantly enhance specificity by mimicking the comprehensive work-up performed by dermatologists.
- 3) **Stain Normalization and Augmentation:** To ensure the model generalizes across different laboratories and scanners, we will implement rigorous stain normalization techniques (e.g., Macenko or Vahadane methods) in the preprocessing pipeline. This will mitigate the impact of color variations in H&E staining that can currently confound the classifier.
- 4) **Explainable AI (XAI):** To facilitate clinical adoption, the "black box" nature of the model must be addressed. We will develop interpretability modules, such as attention maps or Grad-CAM visualizations, to highlight the specific histological features (e.g., atypical lymphocytes, Pautrier's microabscesses) driving the model's predictions, thereby building trust with pathologists.

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VII. FUTURE WORK

To evolve this framework from a promising prototype into a robust clinical decision-support system, future research will address four critical limitations:

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